Emmett Collings

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Education

University of Calgary September 2019 – December 2022
Bachelor of Science, Major in Computer Science *Calgary, AB*

- GPA: 3.8/4.0, focus in theoretical computer science
- Elective Highlights: Computer Security, Distributed Algorithms, Randomized Algorithms, Compiler Construction, Complexity Theory, Network Analysis, Retrogames

Experience

Software Developer - Safety Critical Systems May 2023 – June 2025
Hexagon A&P *Calgary, AB*

- One of the original developers of a greenfield C++ application for monitoring the integrity of GNSS corrections used in autonomous driving technologies
- Implemented application components spanning data processing, inter-application communication, input/output handling, and error handling. Implemented a decoder for the application's primary custom binary data format and contributed to its encoder, developing an in-house bit-array library to support bit-level data manipulation.
- Developed detailed software requirements (ASPICE SWE.1) from established system-level concepts and collaborated on software architecture design, serving as the primary author of the architecture document (SWE.2). Participated in internal and external ASPICE and ISO 26262 audits.
- Performed unit and integration testing and developed CI tooling for automated test execution and code coverage measurement.
- Primary developer of the application's CMake build scripts, contributed to maintaining the company's Python-based build tooling, and contributed to optimizing Jenkins CI pipelines.

C++ Software Developer May 2022 – August 2022
Nobal Technologies *Calgary, AB*

- Developed and maintained software for smart mirror devices using C++, Qt, and QML.
- Played a major role in designing and implementing a hospitality application as part of a rewrite of an existing codebase.
- Integrated third-party services to support hotel functionality, including check-in/out, ride-sharing, and room service.

Software Developer Intern May 2021 – August 2021
Neo Financial *Calgary, AB*

- Developed an internal code coverage tracking solution consisting of a Github Action and web application.
- Built a developer contribution tracking application to monitor code reviews, merged pull requests, and other engineering contributions.
- Both of these tools were adopted in every repository at Neo Financial.

Technical Skills

Languages: C++, C, Python
Testing: Cantata, Google Test, Catch2
Tools: Git, CMake, Perforce, Jenkins
Platforms: Linux, Windows
Standards: ASPICE, ISO 26262, MISRA C++

Research & Projects

Bubbie-VI | *6502 Assembly* September 2022 – December 2022

- Developed a game in 6502 assembly for the Commodore VIC-20
- Operated with strict resource constraints (3.5K of memory available on the VIC-20)
  - Designed custom data compression schemes to minimize memory usage
  - Uses self modification on code and data to increase space efficiency further

NHL Linemates Network | *Python, Jupyter* September 2022 – December 2022

- Analyzed NHL player relationships using network science and statistical techniques.
- Applied community detection and degree centrality to investigate player interactions and identify players who positively influence their linemates.
- Results are summarized in an academic-style research paper (available on GitHub)

Web Proxy | *C, socket programming* January 2022

- Developed an HTTP proxy in C using socket programming.
- Intercepted HTTP responses and modified their content6s, replacing JPEG images and plaintext with custom content.

Interests & Awards

Interests

- Reading (classics, non-fiction)
- Sports and outdoor activities (climbing, volleyball, badminton, swimming, and backpacking)

Awards

- Hexagon A&P High Five - Given to recognize employees internally (2025)
- Deans List (2018, W2020, W2021)